CHAPTER 5 EVIDENCE PACKAGE: IMPLEMENTATION, TESTING AND EVALUATION

> **What this file is.** The Chapter 5 entry point of the nine-file handover set: the implementation and test facts a writer may assert, their evidence, and the exact numbers with dates. Repository `student-beneficiary-system`, branch `fix/sbms-completion-work`, commit `d6b5032`.
>
> **Full detail:** `CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION.md` (§5.1–§5.7), `SBMS_TEST_EVIDENCE_CATALOGUE.md` (every artefact, command and date), `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` (features, artifact, metrics), `SBMS_DEPLOYMENT_AND_OPERATIONS_GUIDE.md` (environments, pipeline).

---

1. Implementation environment (evidence, not recollection)

Item	Value	Status
Operating system	Windows 11 (win32)	Recorded
Node.js / npm	v22.11.0 / 10.9.0	Recorded
Python	3.11.9 in a service virtual environment	Recorded
PostgreSQL	17.11 client and server	Recorded
Docker	CLI 29.9.2; engine **not running** here, image built by CI	Recorded
Browser automation	Headless Chrome over the DevTools Protocol from Node 22's `WebSocket`	Recorded
Hardware specification	**Not recorded**; logs show 7.4 GB RAM with about 0.5 GB free	**Blocking gap G-01**
Test database	Disposable `sbms_test` on loopback, role `sbms_test_runner`, guarded by `testDatabaseGuard.js`	Recorded

The hardware gap must be written as a gap. It is honest to say the workstation was not captured, that memory was observed at 7.4 GB with about 0.5 GB free during sweeps, and that one sweep completed 149 or 150 of 151 tests under memory pressure (G-02). It is not honest to invent a processor, memory size or storage type.

2. Test results: the only counts that may be quoted

Level	Command	Result	Date
Backend Jest, whole tree	`npx jest --runInBand tests`	**24 suites / 151 tests pass**	2026-11-01
Backend Jest, earlier generation	`npm test`	21 suites / 128 tests pass	2026-10-31
Backend native, in-process	`node --test` over seven files	**43/43 pass**	2026-10-31
Backend native, out-of-process	`node --test test/api.smoke.test.js`	**3/3 pass**	2026-10-31
Frontend production build	`npm.cmd --prefix frontend run build`	**exit 0**, 2 006 modules, 10.11 s	2026-11-01
ML service	`unittest discover -s ml-service/tests`	**33/33 pass**	2026-10-31
ML research helper	`unittest discover -s experiments/sbms_dataset`	**7/7 pass**	2026-10-31
Browser: registration and review	`node scripts/browser-smoke.mjs`	**23/23 pass**	2026-10-30
Browser: QR page	`node scripts/qr-smoke.mjs`	**9/9 pass**	2026-10-30
Browser: registration fields, live deployment	`node scripts/registration-fields-smoke.mjs`	**11/11 pass**	2026-11-01
Database restore	`pg_dump` then `pg_restore` into an independent database	Pass, digests compared	2026-10-31
Phone-camera QR scan	manual	**Not run**	—

Level	Command	Result	Date
Browser check of `/verify-certificate`	no harness exists	**Not run**	—
User acceptance	—	**Not conducted**	—
Load or scale test	—	**Not run**	—
Docker build in this environment	—	**Not run** (engine unavailable)	—

Two rules, both mandatory. A count is never quoted without its date, because the suite grew from 21/128 to 24/151 between 2026-10-31 and 2026-11-01. And counts are never summed across levels: 151 Jest tests, 46 native tests, 43 browser checks, 40 Python tests are four separate positions, not a total of 280.

`scripts/admin-reveal-smoke.mjs` exists but was **not** re-run (the seeded administrator password had drifted). It is listed as not run, not as passed and not as skipped.

3. Implementation by feature: what the code does

Feature	Implementation	Tests
Authentication and RBAC	Two sign-in endpoints, JWT, role re-read, user administration	4 + 6 + 1
Registration and validation	Zod schema, five-school constraint, phone and identifier normalisation	9
Duplicate screening	Advisory lock, whole-registry comparison, Python subprocess, `top-1` aggregation, provenance	15 + 13 + 8 + 6 + 1 + 4
Duplicate review	Masked queue, audited reveal, decision with reason, terminal states	9
Eligibility scoring	Five weighted factors, four categories, persisted result	Route and validation only
Award lifecycle	Submission, verification with evidence, approval and rejection with reasons	5
Payments and reconciliation	Issue, clearance with evidence, officer-assisted acknowledgement, discrepancy detection, exceptions	4 + 4 + 4
Student QR identification	32 random bytes, fragment token, existence-only endpoint, dedicated limiter	10 + 4
Award certificates	Snapshot row, reference, token, A4 landscape PDF, revocation, `not_current`	12 + 5
Dashboard, reports, audit, settings	Totals and trend, date-range PDF, audit trail, three cut-offs	6 (authorisation); content by inspection
Operations	Health, readiness, start-up guards, migration ledger	1 + 5 + migration unit tests

4. Machine-learning evaluation: the four numbers that are defensible

Item	Value	Basis
Model	XGBoost over six similarity features	`ml-service/model/`, `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` §5
Live cut-off	0.50: precision 0.9204, recall 0.8927, F1 0.9063, miss 10.73%	Synthetic test partition; operating-point script output
Hard-negative false-positive rate at 0.50	7.72%	Same partition; the number to quote for "how often a wrong pair is flagged"
Independent per-pair baseline	0.35% of 2 000 synthetic unrelated pairs	`docs/ml-evaluation-baseline-2026-10.md`

The 7.72% and 0.35% answer **different questions** (one is the operating point on a constructed hard-negative set, the other a base rate on unrelated pairs). Quote both or neither; quoting one alone misleads. Ranking quality: ROC-AUC 0.9685, PR-AUC 0.9727. Every figure is synthetic, so no real-world accuracy claim is available.

The distinction the chapter must preserve: these are pair-level classification metrics on generated data. They are not beneficiary-level duplicate-detection performance and they do not survive contact with real records until

adjudicated labels exist.

Research-only artifact: `ml-service/model/research/rf_attachment_style_candidate.joblib` is a ten-feature Random Forest candidate. It is not the production model, and its metrics must not be mixed with the six-feature model's (§5.3.4, `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` §11).

5. Defects: six with a fix and a test, others pointed to

#	Defect	Resolution	Validation
1	Duplicate sweep could double-flag under concurrency	Advisory lock around the sweep	Concurrency tests
2	Reveal endpoint leaked full identifiers before adjudication	Masked by default; reveal audited	Route tests
3	QR token in the path was logged	Token moved to the URL fragment	Route and browser tests
4	Certificate PDF ignored `not_current` awards	Snapshot row plus explicit state check	12 route tests
5	Readiness endpoint returned healthy with a stale queue	Queue freshness check added	Readiness tests
6	Payment recorded without verification state	Database-level gate in addition to the API check	Integration tests

The wider defect history — date-of-birth feature skew, registration 500, the deployed `/api/verify` 404, the queue-closes-on-open bug and the readiness false pass — is recorded in `docs/SBMS_PROJECT_HANDOVER_2026-10.md` §13.2–§13.4 and `docs/azure-deployment-log-2026-10-31.md` §7–§8. Chapter 5 §5.4.7 scopes its table to the six and points to these sources rather than implying completeness.

6. Evaluation: what the evidence supports

Objective (Chapter 1)	Extent achieved	Limitation that must be stated
(i) Investigate existing registration and tracking challenges	Achieved as literature and workflow analysis	No field study, no interviews recorded
(ii) Design architecture, schema and ML pipeline	Achieved and implemented	Classifier deviates from the proposal (G-21)
(iii) Implement registration, unique identifiers, QR and RBAC	Achieved	Physical QR scan not exercised (G-11)
(iv) Validate, test and evaluate	Partly achieved	No real data, no user acceptance, no load test

What the tests establish: the delivered behaviour matches the specified behaviour at route and workflow level, across 151 backend tests, 43 browser checks and 40 Python tests, with a reproducible command for each. What they do not establish: fitness for a real bursary office, accuracy on real records, usability, scale, or production assurance.

7. Limits of the evidence itself

Limit	Consequence for the chapter
Results are recorded from runs on one workstation, not from CI for every suite	The chapter states the command, environment and date for each count
Two test generations exist (21/128 and 24/151)	Only the dated, current generation is quoted; the older one is footnoted if mentioned at all
One sweep was incomplete under memory pressure	The stability caveat appears wherever the 151 figure appears
Browser checks depend on a seeded account; one harness was not re-run	The chapter says so rather than implying a full browser sweep
No user acceptance, no load test, no phone capture, no production assurance	Each is named as a gap, with the gap ID

8. Open items affecting Chapter 5

Item	Effect	Owner
G-01 hardware specification	§5.2.1 cannot be completed	Author
G-06 user-acceptance position	§5.5.6 stays empty; NFR-14 stays partial	Author
G-11 phone-camera scan	The identification mechanism rests on a headless-browser check	Author
G-02 149/150-of-151 instability	Every 151 figure needs the caveat	Author
G-25 two test-count generations	A count without a date is misleading	Author

9. Related files

`CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION.md` (the chapter), `SBMS_TEST_EVIDENCE_CATALOGUE.md` (test inventory), `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` (ML evidence), `MASTER_EVIDENCE_REGISTER.md` (evidence IDs).